

iBot Club Workshop Phase

Task Set 0: Circuit Fundamentals

Coordinator Team

February 2026

1 Overview & Motivation

Before working with the Arduino Uno R3, it is strictly required to master breadboard power distribution. **Incorrect wiring on a breadboard is the #1 cause of damaged Arduinos.** This task focuses on understanding power rails and preventing short circuits using a safe battery source before moving to microcontroller power.

2 Component Locations

Hardware for this task is stored in specific locations to facilitate the shared pool system:

- **Batteries (4x AA) & Holders:** Located on the **Centre Table**.
- **LEDs, Resistors, & Push Buttons:** Middle drawer of the white cupboard.
- **Jumper Wires:** In the box on the **Open Shelf**.

3 Task 0.1: The Power Rail (LED Glow)

Objective

Learn to utilize the power rails on a breadboard to distribute power and ground.

Components Required

- 1x Battery Holder with 4x AA Batteries
- 1x Breadboard
- 1x LED (Any color)
- 1x 330 Ω Resistor
- Jumper Wires (Male-to-Male)

Instructions

1. Connect the Battery Holder wires to the **Power Rails** (the long horizontal lines usually marked Red/Blue or +/-) of the breadboard.
2. Insert the LED and the Resistor into the main component area (terminal strips).
3. Use jumper wires to connect the components to the Power Rails.
4. **Requirement:** You must use the power rails. Do not plug the battery directly into the same column as the LED.
5. **Check:** The LED should glow continuously.

4 Task 0.2: Manual Control (Push Button)

Objective

Understand the internal structure of a 4-terminal tactile switch and implement a series control circuit.

Theory: The 4-Terminal Switch

The push buttons provided have 4 legs.

- Internally, pins are connected in pairs (Leg 1 is connected to Leg 2; Leg 3 is connected to Leg 4).
- When pressed, the two pairs are bridged, completing the circuit.
- *Warning:* Connecting the wrong legs can result in a permanently closed circuit or a short.

Components Required

- Same components as Task 0.1
- 1x Push Button (from drawer)

Instructions

1. Modify the circuit from Task 0.1.
2. Place the push button on the breadboard, bridging the central ravine (the gap in the middle) or ensuring legs are in separate columns.
3. Connect the Push Button in **Series** with the LED and Resistor.
4. Ensure the entire circuit is physically on the breadboard, powered via the rails.
5. **Check:** The LED should glow **ONLY** when the button is pressed.

5 Cleanup & Return Policy (Mandatory)

Since hardware is shared among all Deputy Coordinators, you must adhere to the following before leaving your slot:

1. **Dismantle:** Completely disconnect all wires and components from the breadboard.
2. **Sort & Return:**
 - **Batteries:** Return to the **Centre Table**.
 - **Components (LEDs, Resistors, Buttons, Breadboard):** Return to the **Middle Drawer**.
 - **Jumpers:** Return to the box on the **Open Shelf**.
3. **Clear Desk:** Ensure the workspace is empty for the next person.

6 Submission

- **GitHub:** Not required for this task.
- **Video:** Upload a video to OneDrive showing the working circuit for Task 0.2 (demonstrate the button press) and briefly explain the current flow. Submit the link via the Google Form.